Inductors

An inductor is a coil of wire that resists the change of current through the coil. An inductor is sometimes called a choke.

Self inductance/ Inductance

When the current through the inductor changes (e.g. increases) a (changing) magnetic flux will be induced by right hand grip rule. This changing magnetic flux also induces a voltage across the inductor by faradays law, and by lenz's law, a current that opposes the changing magnetic flux is induced, causing the inductor to resistance the change of current.

This means that an increase in current does not happen immediately, but gradually. Similarly, a decrease in current does not happen immediately but gradually.

Note that if the input current is always changing, the current through the component and its voltage will also be changing

This property is called self inductance, or inductance.

Inductance has a symbol L and has units Henries (H).

We can find the induced voltage across a inductor as so:

$$v_i = L\frac{di}{dt}$$

One thing to note is that when a current I passes through the current after some time, di/dt= 0, thus the induced voltage is 0, aka voltage across inductor is 0 (essentially a wire now, thus V = 0)

Another way to determine induced voltage is by the change in the induced magnetic flux due to the current flowing through the inductor.

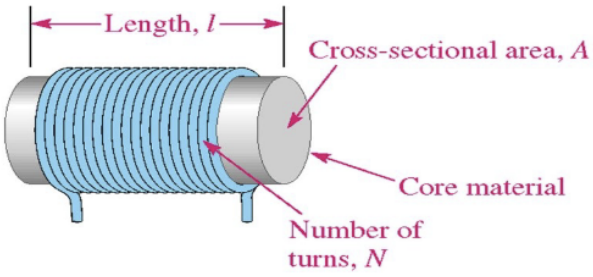
$$v_i = N\frac{d\Phi}{dt}$$

Notice how this is the same as Faradays Law, as the voltage induced by the inductor is due to Faraday's Law.

Inductance is determined by the physical properties of the inductor, by the following:

$$L = \frac{N^2 \mu A}{l}$$

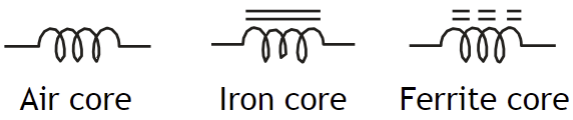
where N is the number of coil turns, mu is the permeability of the core in the inductor, A is the cross sectional area of the coil, and l is the length of the coil(not the wire!!!!)



Types of inductors

Inductors can be sorted by if they have a fixed or variable inductance.

They can also be sorted by the core used, air, iron or ferrite.



Energy stored

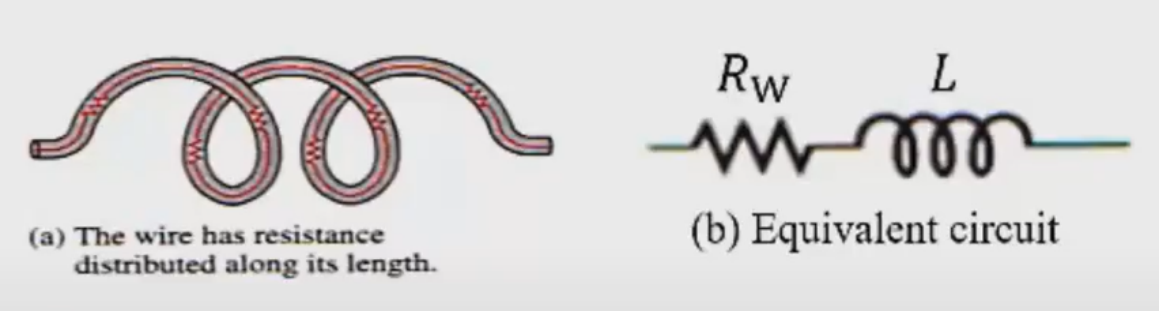
When a inductor creates a magnetic field, it stores energy in the magnetic field. The stored energy can be found as

$$W = \frac{1}{2} L I^2$$

Notice how this formula is similar to energy stored by a capacitor.

Winding Resistance and Capacitance

As an inductor is a coil of wire, it can have its own resistance distributed across the whole length of the coil, known as winding resistance(Rw). We can represent this resistance as a resistor in series with this.

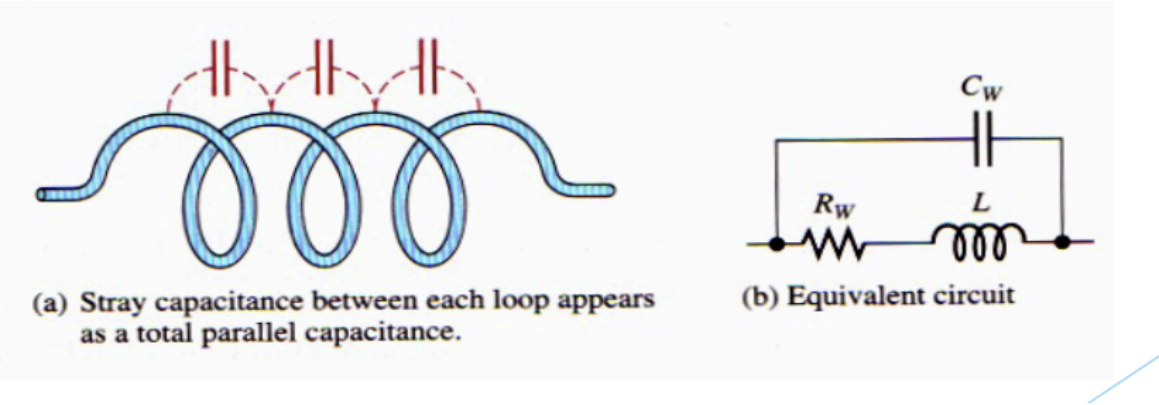


Usually, the resistance is quite small, which can be assumed to be an ideal inductor.

Similarly, a coil of wire can have some stray capacitance, where there is some parallel capacitance between each turn of the inductor.

Usually, this capacitance has no significant effect. However, with high frequency AC, it becomes more important.

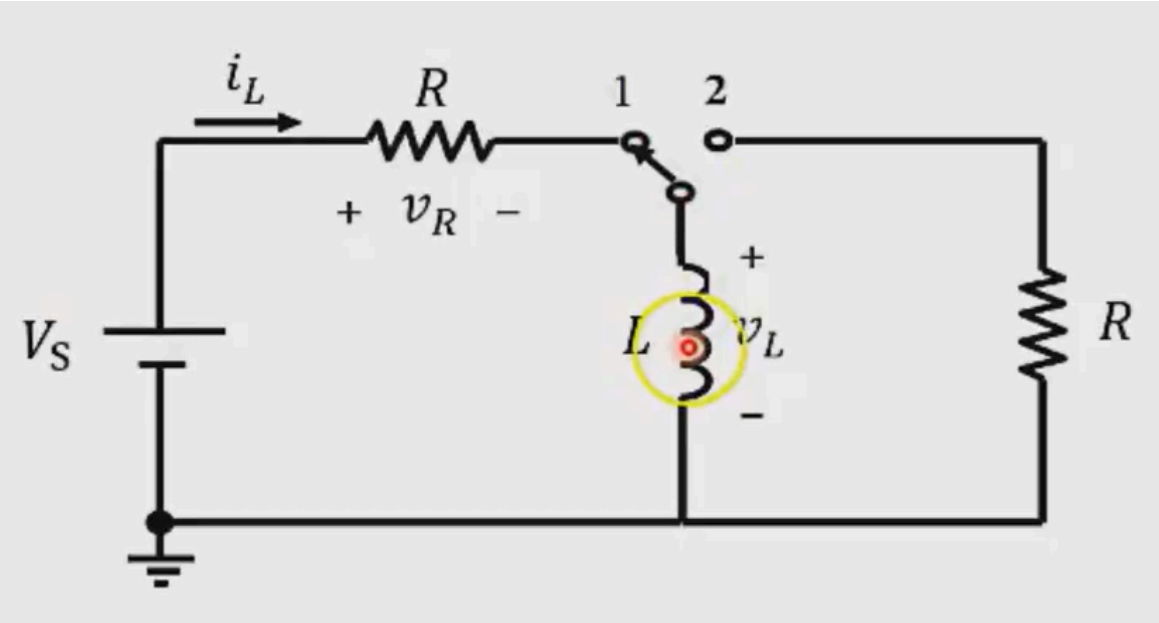
We can represent the winding capacitance similarly:



Note that the capacitor is now in parallel with the inductor.

Charging Inductors

Similarly to capacitors, inductors can be charged in a similar circuit.



Note how the voltage across the inductor is opposite the power source (opposes voltage, recall from lenz's law)

When the switch is just closed, the inductor resists the instant surge of current, thus I=0. Thus the current through the whole circuit (and other components) will also be 0. By KVL however, the voltage across the inductor will be Vs.

Thus, at t = 0 the inductor acts as a open circuit (opposite of capacitor)

As time passes, the current will start to increase exponentially while the voltage will decrease exponentially, based on the time constant τ.

The time constant is found by:

$$\tau = \frac{L}{R}$$

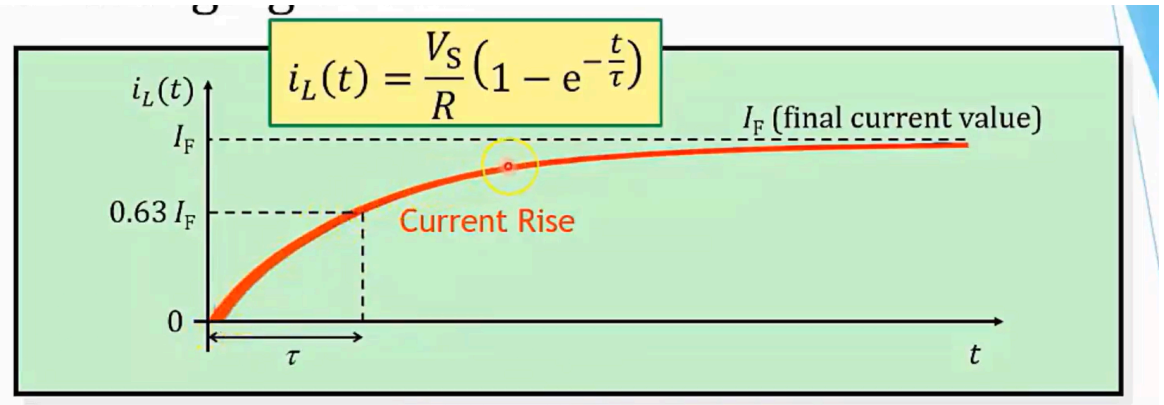
Similarly, after τ seconds, the current has reached 63% of the total current and after 5τ seconds, the inductor has reached the maximum current.

At that point, the inductor is essentially a wire (as stated earlier), thus it can be considered a short circuit.

The relationship of current and voltage of the inductor against time can be found as so:

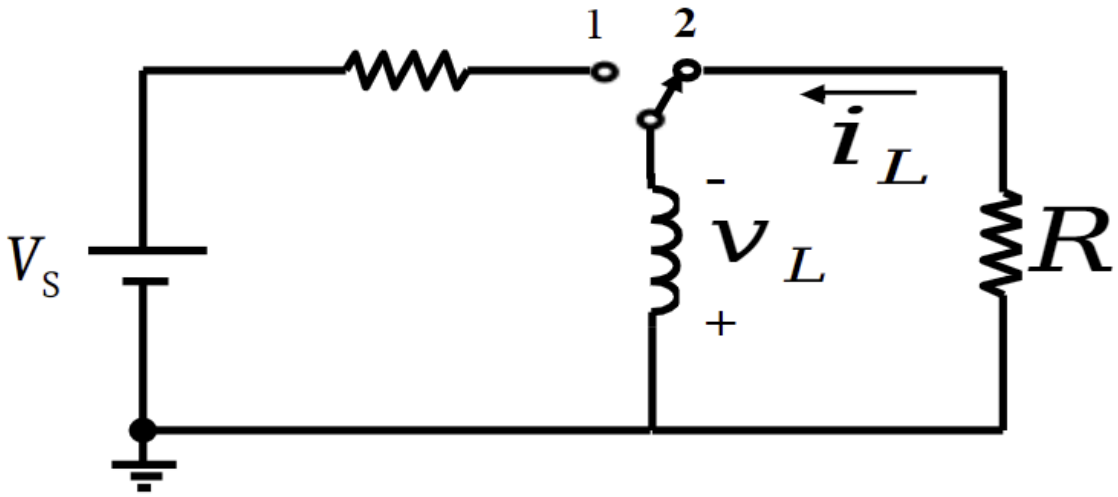
$$i_L(t) = \frac{V_s}{R}(1 - e^{-\frac{t}{\tau}})$$

$$v_L(t) = V_s e^{-\frac{t}{\tau}}$$



Notice how the relationships are similar to capacitors, however relating to different quantities.

Discharging Inductors



The instant the switch is flipped, the inductor still holds onto the current and keeps the same voltage polarity.

As it discharges, the current decreases exponentially similarly as so:

$$i_L(t) = \frac{V_s}{R} e^{-\frac{t}{\tau}}$$

Similar concepts apply, where the current drops by 63% after τ seconds, and almost all of the current is discharged after 5τ seconds.

Circuit Analysis with Inductors

For n inductors connected in series,

$$L_T = \sum_{k=1}^{n} L_k = L_1 + L_2 + ... + L_n$$

For n inductors in parallel,

$$L_T = \left(\sum_{k=1}^{n} L_k^{-1} \right)^{-1} = \left(\frac{1}{L_1} + \frac{1}{L_2} + ... + \frac{1}{L_k} \right)^{-1}$$

aka same as resistors.